INTELLIGENT STREAM GUIDANCE SYSTEM FOR O/L STUDENTS

25-26j-336

Project Proposal Report

Bandara R M M K T – IT22897008

B.Sc. (Hons) Degree in Information Technology Specializing in Information Technology

Department of Computer Science

Sri Lanka Institute of Information Technology Sri Lanka

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Supervised by Mrs.
Co-supervised by Ms.

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DECLARATION OF THE CANDIDATE & SUPERVISOR

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Bandara R.M.M.K.T	IT22897008	

The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor (Ms. Jenny Krishara)	2025/08/30
Signature of the Co supervisor (Ms. Poorna Panduwawala)	30.08.2025

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Finally, we thank the academic community and researchers whose prior work in educational data mining and machine learning formed the foundation for our research approach.

ABSTRACT

Academic stream selection is one of the most critical decisions for O/L students in Sri Lanka, yet determining the optimal A/L stream remains challenging due to the complex interaction of academic performance patterns, individual aptitudes, and evolving educational requirements. Accurate stream recommendation is crucial for maximizing student success rates, reducing dropout rates, and ensuring efficient allocation of educational resources. Traditionally, stream selection relies on manual counseling sessions or basic grade cutoffs, which can be subjective, inconsistent, and fail to capture individual learning patterns and potential.

The goal of this research is to develop an intelligent machine learning-based system that can predict optimal academic streams and provide personalized guidance for O/L students without requiring expensive assessment tools. By utilizing historical term test results from grades 9-11, combined with structured quiz responses and performance analytics, the system predicts stream suitability and identifies areas for academic improvement. A key innovation of this approach is the integration of Natural Language Processing (NLP) techniques, particularly BERT-based models, to analyze structured quiz responses and assess conceptual understanding across different subject areas.

Based on the predicted stream compatibility, the system provides personalized recommendations for improvement areas, suggests relevant study resources, and offers alternative pathways when preferred streams may not be suitable. The application incorporates Explainable AI methods to clearly identify how academic performance patterns and quiz responses influence recommendations, providing transparent insights for students, parents, and educators. Additionally, the system tracks student progress over time and adjusts recommendations based on O/L results and changing performance trends.

This solution offers a more systematic, data-driven, and accessible alternative to traditional stream selection methods, enabling better educational decision-making, reducing academic mismatches, and supporting students in achieving their full potential in higher secondary education.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from Transformers
O/L	Ordinary Level
A/L	Advanced Level
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
F1-score	F1 Measure
XGBoost	Extreme Gradient Boosting
API	Application Programming Interface
UI	User Interface
CSV	Comma Separated Values
PDF	Portable Document Format

1 INTRODUCTION

1.1 Background and Literature Survey

1.1.1 Background

Academic stream selection represents one of the most critical educational decisions for students in Sri Lanka's secondary education system. The choice of Advanced Level (A/L) stream whether Physical Science, Biological Science, Commerce, Arts, or Technology significantly influences a student's future career prospects, university admission opportunities, and long-term academic success. This decision is influenced by a complex set of factors, including individual academic performance patterns, subject-specific aptitudes, cognitive abilities, career aspirations, and evolving market demands for different professional fields.

Traditionally, stream selection and academic guidance rely on manual counseling sessions conducted by teachers or career advisors, often based on simple grade cutoffs from Ordinary Level (O/L) examinations. While these conventional methods provide some guidance, they are often inadequate, subjective, and fail to capture the nuanced patterns of individual learning trajectories and potential. Inappropriate stream selections frequently lead to poor academic performance in A/L studies, increased dropout rates, and suboptimal utilization of educational resources.

Academic performance prediction in secondary education presents additional challenges. Success in A/L streams depends not only on O/L results but also on sustained performance patterns observed throughout grades 9-11, conceptual understanding across subject areas, and individual learning preferences. Without comprehensive analysis of these factors, students often make uninformed decisions that result in academic struggles, career dissatisfaction, and inefficient allocation of educational opportunities.

In recent years, advances in artificial intelligence (AI) and machine learning (ML) have opened new possibilities for more accurate and data-driven educational guidance systems. Historical academic records, term test results, and student performance datasets can be used to train ML models capable of predicting stream compatibility with greater accuracy than traditional methods. Moreover, the integration of Natural Language Processing (NLP) techniques allows these systems to analyze structured responses and assess conceptual understanding beyond simple numerical grades.

By combining academic performance prediction with personalized aptitude assessment and explainable recommendations, educators can guide students more effectively toward appropriate academic pathways. The inclusion of continuous monitoring and adaptive feedback mechanisms enables the system to refine recommendations based on evolving student performance and changing educational landscapes.

The need for systematic, transparent, and accessible academic guidance has driven the integration of machine learning, natural language processing, and educational data analytics into a unified approach for stream recommendation and academic planning, offering a modern, scalable solution for Sri Lankan secondary education.

1.1.2 Literature Survey

In recent years, machine learning (ML) and recommender systems have been increasingly applied in education to support student performance prediction and academic guidance.

A Sri Lankan study developed SubjectStreamPredict, a machine learning based system that recommends suitable A/L subject streams for students using O/L results and aptitude data. The system compared Decision Trees, Random Forest, K-Nearest Neighbor, and Support Vector Machines, with Random Forest achieving around 70% accuracy [1].

Further work in Sri Lanka examined the factors influencing A/L stream selection, particularly the low enrollment in the Technology stream. Findings revealed that students' decisions were shaped by career opportunities, limited awareness, lack of guidance at the O/L stage, and insufficient encouragement from parents and teachers [2].

At an international level, recommender systems in education have been widely studied. A systematic review categorized 98 systems into collaborative filtering, content-based, and hybrid models, noting a growing trend of integrating ML techniques for personalized learning [3].

More recent research has emphasized fairness and adaptability in recommendation. A hybrid recommendation framework for K-12 education integrated graph modeling and matrix factorization while incorporating fairness to reduce bias in guidance [4]. Similarly, DK-PRACTICE, an adaptive learning platform, implemented diagnostic assessments, real-time feedback, and personalized learning material recommendations using student knowledge state modeling [5].

These studies highlight the growing role of ML and adaptive systems in education. However, gaps remain in integrating multi-year academic performance, aptitude assessments, and localized guidance for A/L stream selection in Sri Lanka.

1.2 Research Gap

Based on the reviewed literature, several key gaps emerge:

- Longitudinal Analysis: SubjectStreamPredict [1] considered O/L data only, without incorporating multi-year term-test performance (Grades 9-11).

- NLP/Aptitude Integration: While adaptive testing systems like DK-PRACTICE [5] provided real-time assessments, they were not applied to stream-specific recommendations in the Sri Lankan education system.

- Limited Real-Time Guidance: Most recommended systems [3], [4] offer resource suggestions but do not dynamically adapt based on student exam performance and preferences.

- Sri Lankan Context: Studies in Sri Lanka [1], [2] examined prediction models and social factors separately but did not merge predictive analytics with actionable guidance.

- Unified System Gap: No existing work integrates student preferences, longitudinal academic results, aptitude-based quizzes, and improvement-oriented recommendations into a single intelligent platform tailored to Sri Lanka's A/L stream selection.

Research Paper	Longitudinal Analysis (Grades 9-11)	NLP-based Aptitude Assessment	Stream-specific Recommendations	Real-time Guidance	Sri Lankan Context
Fernando (2022) [1]	✗	✗	✓	✗	✓
Navaneethakrishnan & Nithiyananthan (2024) [2]	✗	✗	✗	✗	✓
Tarus et al. (2021) [3]	✗	✗	✗	✗	✗
Cai et al. (2025) [4]	✗	✗	✗	✓	✗
Chen et al. (2024) [5]	✗	✓	✗	✓	✗

Table 1:Research Gap

2 RESEARCH PROBLEM

For effective academic guidance and career pathway optimization in Sri Lanka's secondary education system, accurate stream recommendation is crucial for student success and educational resource allocation. However, there are significant obstacles that existing methods of academic guidance must overcome, such as reliance on subjective counseling approaches, limited integration of longitudinal performance data, lack of systematic aptitude assessment, and absence of interpretable decision-making frameworks. The development of comprehensive, data-driven, and transparent guidance systems that can be widely adopted in educational institutions is hampered by these limitations. The following is a summary of the main concerns that must be addressed in the research problem:

Dependence on Subjective Assessment and Manual Counseling

Many existing stream recommendation approaches rely on traditional counseling sessions, teacher opinions, or simple grade cutoffs from O/L examinations to guide students toward A/L streams. While these methods incorporate human expertise, they are often inconsistent, subjective, and heavily dependent on the experience and availability of qualified counselors. For students in rural schools or under-resourced educational institutions, such personalized guidance is often inaccessible or inadequate. A scalable solution must therefore avoid dependence on manual assessment and instead leverage systematic analysis of academic performance patterns and objective evaluation methods.

Narrow Focus on Final Grades Without Longitudinal Analysis

Most traditional guidance approaches are limited to considering only final O/L examination results or recent academic performance. However, stream compatibility assessment requires understanding of consistent performance patterns, learning trajectories, and subject-specific strengths developed over multiple years. Current methods often fail to analyze the progression of student capabilities from grades 9-11 or identify underlying aptitudes that may not be reflected in a single examination. Without comprehensive longitudinal analysis, students may be guided toward inappropriate streams despite having demonstrated consistent strengths or improvement patterns in specific subject areas.

Lack of Systematic Aptitude Assessment Beyond Numerical Grades

Although traditional academic records provide quantitative measures of student performance, they often fail to capture conceptual understanding, critical thinking abilities, and subject-specific aptitudes that are crucial for A/L success. Current guidance systems rarely incorporate systematic evaluation of students' ability to apply knowledge, solve complex problems, or demonstrate deep understanding of core concepts. This limitation is particularly problematic for streams like Physical Science or Biological Science, where conceptual mastery is more predictive of success than memorization-based performance.

Absence of Interpretable and Transparent Decision-Making

Even when data-driven approaches are attempted in educational guidance, the decision-making process often remains opaque to students, parents, and educators. Students may receive stream recommendations without understanding the underlying rationale or the specific areas where improvement is needed. This lack of transparency limits trust in the guidance system and fails to provide actionable insights that could help students strengthen their preparation for chosen streams or consider alternative pathways.

Underutilization of Comprehensive Academic Data

While extensive academic data exists in educational institutions - including term test results, subject-wise performance patterns, attendance records, and assignment evaluations - most guidance systems do not systematically leverage this information. Instead, they rely primarily on final examination scores or teacher impressions, which may not capture the full picture of student capabilities and potential. There is significant opportunity to integrate multiple data sources to create more accurate and comprehensive student profiles for stream recommendation.

Need for Continuous Monitoring and Adaptive Guidance

Academic guidance is often provided as a one-time consultation before A/L stream selection, which does not account for evolving student performance, changing interests, or academic improvement over time. Students require systems that can provide ongoing assessment and adapt recommendations based on new performance data and changing circumstances. Without continuous monitoring, guidance systems fail to identify students who may have initially been directed toward suboptimal streams but have since demonstrated improved capabilities in other areas.

Potential of Machine Learning and NLP for Comprehensive Assessment

These challenges can be addressed through advanced machine learning techniques combined with Natural Language Processing (NLP) for comprehensive student assessment. By applying algorithms such as ensemble methods for performance prediction and BERT-based models for analyzing structured responses, the system can provide both quantitative performance analysis and qualitative aptitude assessment. This approach enables transparent explanations of recommendations while providing actionable insights to help students optimize their academic preparation and make informed decisions about their educational pathways.

By addressing these challenges, the proposed research will provide a novel, interpretable, and scalable machine learning-based system for academic stream guidance. Unlike existing approaches, it will integrate longitudinal performance analysis, systematic aptitude assessment, personalized recommendations, and explainable decision-making while avoiding dependence on subjective manual counseling. The system can be easily deployed in secondary schools and adapted to diverse educational contexts, ultimately supporting data-driven academic planning, reducing stream mismatches, and improving student success rates in higher secondary education.

3 OBJECTIVES

3.1 Main Objective

The primary objective of this study is to develop and implement an intelligent machine learning-based system for academic stream recommendation, aptitude assessment, and personalized guidance for O/L students. The system will integrate ensemble learning models trained on longitudinal academic performance data from grades 9-11 and NLP-based assessment of structured quiz responses to generate accurate stream compatibility predictions without relying on expensive assessment tools or subjective counseling methods. In addition, it will provide personalized improvement recommendations and identify specific areas for academic development. By combining these capabilities into a unified platform, the proposed system aims to support data-driven educational decision-making, reduce stream mismatches, and improve student success rates in higher secondary education.

3.2 Specific Objectives

Specific

- Collect academic performance data from term tests (grades 9-11) and O/L results from past examination records.
- Apply preprocessing methods such as data cleaning, normalization, handling missing grades, and feature engineering for academic performance indicators.
- Extract meaningful features such as subject-wise performance patterns, grade trajectories, consistency metrics, and improvement trends.
- Develop and train machine learning models (e.g., Random Forest, XGBoost, ensemble methods) for stream compatibility prediction.
- Implement NLP-based models (BERT) for analyzing structured quiz responses and assessing conceptual understanding.
- Design a transparent recommendation system that provides clear reasoning for stream suggestions.

Measurable

- Evaluate model performance using metrics such as accuracy, precision, recall, and F1-score for stream classification.
- Compare predicted stream recommendations with actual A/L performance outcomes to validate system effectiveness.
- Assess recommendation accuracy against expert counselor evaluations using Cohen's kappa coefficient.
- Measure system usability through user feedback surveys from students, parents, and educational professionals.

Achievable

- Ensure academic dataset preprocessing and feature extraction can be performed using standard computational resources available in educational institutions.
- Train models that achieve acceptable prediction accuracy (>85%) within reasonable computational constraints.
- Confirm that recommendation explanations and improvement suggestions can be understood by students and parents.

Relevant

- Address the critical need for systematic and objective academic guidance in Sri Lankan secondary education.
- Provide educational institutions with data-driven tools for student counseling and resource allocation.
- Contribute to improved educational outcomes by reducing inappropriate stream selections and academic dropouts.

Time-bound

- Collect and preprocess academic datasets within two months.
- Develop and train the initial stream recommendation models within three months.
- Implement NLP-based quiz analysis and integrate with performance prediction within four months.
- Complete transparent recommendation system and conduct user testing within one month of model integration.
- Validate system performance with pilot deployment in selected schools within six months.

4 METHODOLOGY

The proposed methodology integrates multiple stages of data collection, preprocessing, machine learning model development, and natural language processing to deliver accurate academic stream recommendations, aptitude assessment, and personalized guidance for O/L students. The primary objective is to design a reliable and interpretable AI system that supports students, parents, and educators in making informed decisions about academic pathways and career planning. Each stage of the methodology is described below.

Data Collection

The system will utilize comprehensive academic datasets from multiple educational institutions, supplemented with examination board records and historical performance data where available. These datasets will include information such as:

- **Academic performance variables:** Term test scores from grades 9-11, subject-wise performance patterns, grade progression trends, and past O/L examination results from previous student cohorts.
- **Assessment data:** Current user O/L examination results for validation, preferred A/L stream selections, and quiz response data.
- **Student interaction data:** Quiz responses, improvement areas identified, and system usage patterns to enhance personalization.

By leveraging diverse educational data sources, the system will be trained to provide robust recommendations across varying academic contexts and institutional settings.

Preprocessing

Raw academic datasets often contain inconsistencies, missing values, or irrelevant features. To ensure high-quality model input, the following preprocessing steps will be implemented:

- **Data Cleaning:** Handling missing grades using appropriate imputation methods (mean, median, or forward-fill for time series) and removal of duplicate or inconsistent entries.
- **Normalization and Scaling:** Standardizing numerical variables (e.g., test scores, grade percentages) to ensure consistency across different grading systems and examination formats.
- **Feature Engineering:** Generating derived features such as grade improvement trends, subject-wise performance ratios, consistency indices, and academic trajectory patterns to capture longitudinal learning behaviors.

- **Outlier Detection:** Identifying and addressing anomalies in academic records or assessment scores to avoid biased recommendations.

Model Development

The AI system will integrate multiple machine learning components to predict stream compatibility, assess aptitude, and provide personalized guidance.

- **Stream Prediction Model:**
  Machine learning algorithms such as Random Forest, XGBoost, or ensemble methods will be trained to predict optimal A/L stream selection. Input features will include longitudinal academic performance, subject-specific strengths, and grade progression patterns from historical data.
- **NLP-based Aptitude Assessment:**
  BERT-based natural language processing models will analyze structured quiz responses to evaluate conceptual understanding, critical thinking abilities, and subject-specific competencies beyond numerical grades.
- **Recommendation Validation System:**
  Based on collected user O/L results, the system will continuously validate and refine its predictions to improve accuracy for future recommendations.
- **Model Training and Validation:**
  A supervised learning approach will be used, where the system will be trained on labeled data (e.g., historical student records and subsequent A/L outcomes). Performance will be evaluated using cross-validation and temporal validation to ensure generalizability across different academic years and institutional contexts.

Natural Language Processing Integration

Advanced NLP techniques will be incorporated to assess qualitative aspects of student learning:

- **Quiz Response Analysis:** BERT models will evaluate structured answers to subject-specific questions, identifying conceptual understanding and knowledge gaps.
- **Semantic Similarity Assessment:** Comparing student responses with model answers to provide partial credit and detailed feedback on conceptual mastery.
- **Learning Pattern Recognition:** Analyzing response patterns to identify individual learning styles and cognitive preferences that influence stream compatibility.

Sequential Assessment Process
The system follows a structured workflow to maximize prediction accuracy:

- **Initial Assessment:** Students input grades 9-11 term test results and preferred A/L stream to receive preliminary recommendations.
- **Targeted Quiz Generation:** System generates personalized quizzes based on initial assessment to evaluate specific competencies.
- **Refined Recommendations:** Quiz results are integrated with academic performance data to provide final stream recommendations and improvement suggestions.
- **Outcome Validation:** User O/L results are collected to validate predictions and continuously improve model performance.

Evaluation and Validation
To ensure reliability and educational effectiveness, the system will be evaluated using multiple criteria:

- **Prediction Accuracy**: Metrics such as accuracy, precision, recall, and F1-score for stream classification performance.
- **Educational Outcome Validation:** Comparison of predicted recommendations with actual A/L performance outcomes to validate system effectiveness.
- **Longitudinal Analysis:** Tracking student success rates in recommended streams over multiple academic cycles.
- **Cross-Institutional Validation:** Testing across different school types and regions to ensure adaptability and fairness.

Practical Integration
The system will be built with educational usability in mind, ensuring seamless integration into existing school guidance workflows:

- **User-Friendly Interface:** Students, parents, and educators can input academic data and receive recommendations through an intuitive web-based platform.
- **Lightweight Deployment:** The model will be designed to run on standard educational institution infrastructure without requiring specialized hardware or extensive technical support.
- **Scalability:** Capable of handling multiple students simultaneously and expanding to accommodate different educational systems and assessment formats.

4.1 System Architecture

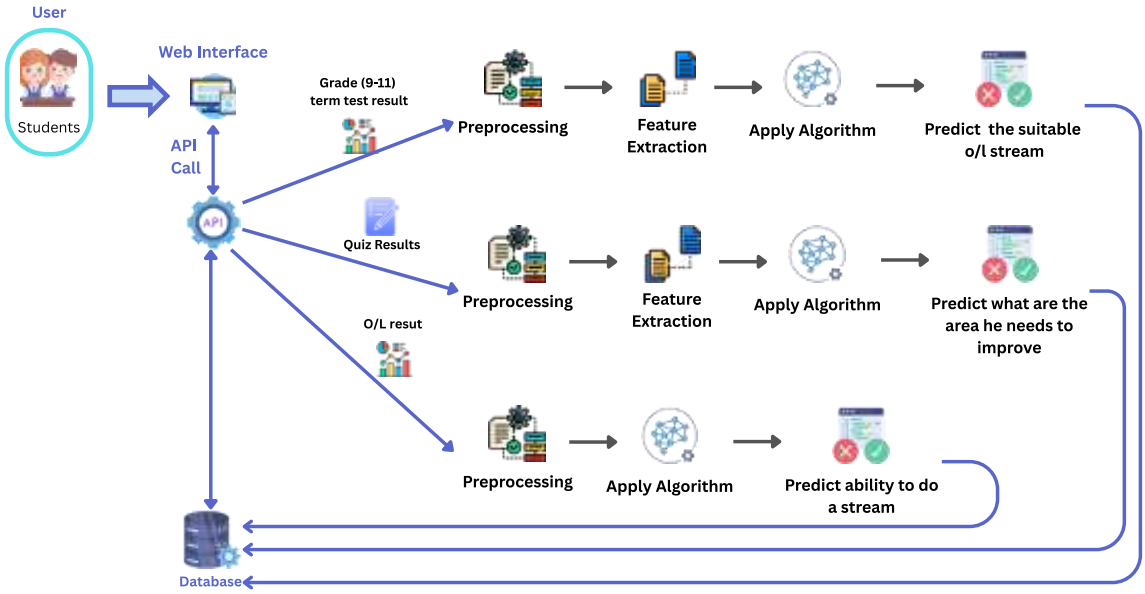

Figure 1:System Diagram

4.1.1 Software solution

The proposed intelligent stream guidance system for O/L students will be developed following the Software Development Life Cycle (SDLC) methodology to ensure structured development, risk management, quality assurance, and timely delivery within budget. Each SDLC phase requirement gathering, planning, system design, implementation, testing, deployment, and maintenance is tailored to guarantee reliability, usability, and practical applicability for students, parents, and educational counselors.

Requirement Gathering

The team will first collaborate with domain experts, including educational counselors, teachers, examination board officials, and academic administrators, to define system requirements. This includes:

- Identifying critical features for stream prediction such as longitudinal academic performance, subject-specific aptitudes, and learning patterns.
- Understanding current guidance practices, counselor workflows, and student decision-making processes.
- Reviewing historical academic datasets and O/L examination records to establish benchmarks for model training and validation.
- Assessing gaps in existing guidance systems and traditional counseling methods to define areas for improvement, such as systematic assessment and personalized recommendations.

Planning

A feasibility study will be conducted to evaluate project scope, resource requirements, and technological considerations. Key steps include:

- Determining suitable machine learning models for stream prediction (e.g., Random Forest, XGBoost) and NLP models for quiz analysis (e.g., BERT).
- Estimating computational resources for model training, including cloud-based services for processing large educational datasets.
- Establishing a project timeline for data collection, preprocessing, model development, testing, and deployment.
- Defining evaluation metrics for accuracy, educational effectiveness, and user satisfaction.

4.2 Domain and Technology

4.2.1 Domain

The system operates in the educational data analytics domain, focusing on academic guidance and career pathway optimization. It provides:

- Accurate prediction of optimal A/L streams using longitudinal academic performance data.
- Assessment of student aptitudes and conceptual understanding through NLP-based quiz analysis.
- Personalized recommendations for academic improvement and study resource allocation.
- Actionable insights for educators via interpretable visualizations, improving student counseling effectiveness.

4.2.2 Technology

Key tools and technologies include:

- Python:Core programming language for data processing, machine learning, and web development.
- scikit-learn/XGBoost/LightGBM: ML frameworks for stream prediction and performance analysis.
- Transformers/BERT: NLP models for quiz response analysis and semantic understanding.
- Pandas/NumPy: Data preprocessing and manipulation of academic records.
- Matplotlib/Seaborn/Plotly: Visualization of academic trends and recommendation insights.
- Flask/Django: Web framework for developing user interface and API endpoints.
- PostgreSQL/MySQL: Database management for storing student data and academic records.
- AWS / Google Cloud: Cloud computing for model training, storage, and deployment.
- Jupyter Notebook: Interactive environment for prototyping and experimentation.
- GitLab/GitHub: Version control and collaborative development.

5 PROJECT REQUIREMENTS

5.1 Functional Requirements

- **Data Acquisition:** Collect data from multiple sources, including historical O/L examination records, term test results from grades 9-11, and student academic profiles. The system should allow batch imports of academic data and updates from examination boards and educational institutions.
- **Data Preprocessing and Feature Engineering:** Clean, normalize, and scale academic datasets. Generate derived features like grade improvement trends, subject-wise performance ratios, consistency indices, and academic trajectory patterns. Handle missing grades and remove anomalous entries.
- **Machine Learning Model Training and Testing:** Train models for stream prediction, aptitude assessment, and performance forecasting. Evaluate multiple algorithms to optimize for accuracy, educational relevance, and computational efficiency.
- **NLP-based Quiz Analysis:** Implement BERT models to analyze structured quiz responses, assess conceptual understanding, and provide detailed feedback on subject-specific competencies beyond numerical grades.
- **Real-Time Assessment and Recommendations:** Enable students to input academic data and receive immediate stream recommendations, personalized improvement suggestions, and study resource guidance. Provide interactive dashboards for clear interpretation of results.

5.2 Non-Functional Requirements

- **Reliability**: Accurate predictions across varying school types, grading systems, and regional educational contexts.
- **Performance**: Recommendations must be generated quickly (e.g., within seconds for typical student datasets).
- **Security**: Ensure secure storage and transmission of student data using encryption and access controls.
- **Data Privacy**: Anonymize sensitive student information and restrict access to authorized educational personnel only.

5.3 System Requirements

Hardware Requirements:
Standard desktop or laptop for data preprocessing and model inference.
Optional cloud environment for training on large educational datasets.
Lightweight deployment possible on school computers or tablets for counselors and administrators.

Software Requirements:
Python with necessary ML and NLP processing libraries.
Jupyter Notebook or similar IDE for model development and experimentation.
Web framework (Flask/Django) for user interface and API development.

User Interface Requirements:
Display stream recommendations, aptitude assessments, and improvement suggestions in an easy-to-read format.
Include interactive visualizations and academic progress tracking.
Allow export of guidance reports in PDF format for student records and counseling sessions.

Performance Requirements:
Achieve high prediction accuracy (e.g., F1-score > 0.85 for stream classification).
Provide low-latency results for practical use during counseling sessions.

Security and Privacy Requirements:
Anonymize sensitive student academic data and personal information.
Implement user authentication and role-based access for educational staff.
Encrypt stored data and communication channels according to educational data protection standards.

Compatibility Requirements:
Compatible with Windows, Linux, and MacOS for broader accessibility in schools.
Support standard data formats (CSV, Excel) and allow integration with existing school management systems.
Modular architecture for future extension, e.g., incorporating additional assessment methods or expanding to other educational levels.

6 TIMELINES

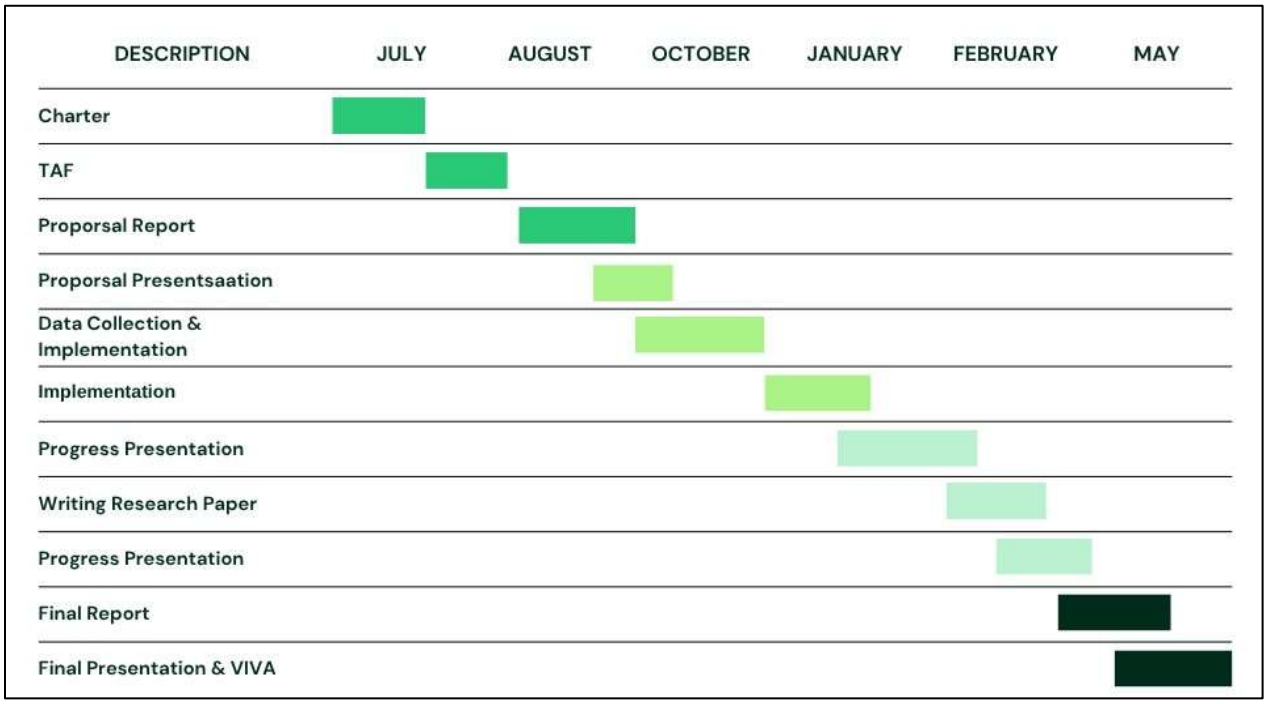

Figure 2:Time Line

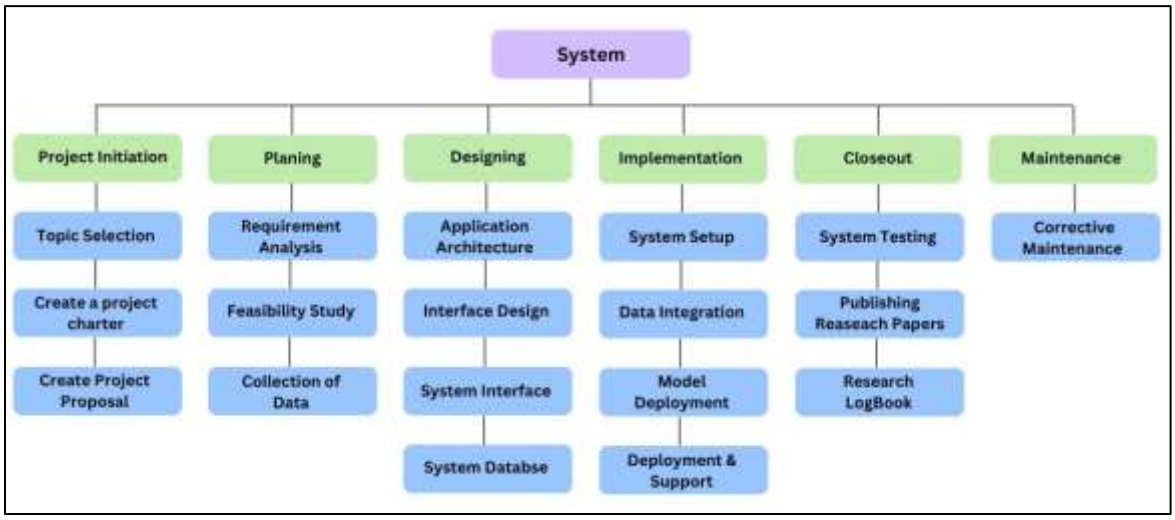

Figure 3:work Breakdown

7 BUDGET AND BUDGET JUSTIFICATION

Item	Description	Estimated Cost (LKR)
Hardware	Personal laptop (already available, no additional cost)	0.00
Software Tools	Python, Jupyter Notebook, Scikit-learn, ReactJS, Flutter, Flask/Django (open-source)	0.00
Database & Storage	MySQL/PostgreSQL (open source) + Firebase (free tier for initial testing)	0.00
Internet & Connectivity	Internet package (research, collaboration, deployment testing)	5,000.00
Testing & Survey Materials	Printing questionnaires, forms, and materials for student/teacher pilot surveys	3,000.00
Miscellaneous Costs	Transport, communication, unexpected expenses during pilot testing	5,000.00
Total Estimated Cost		13,000.00

Table 2:Budget

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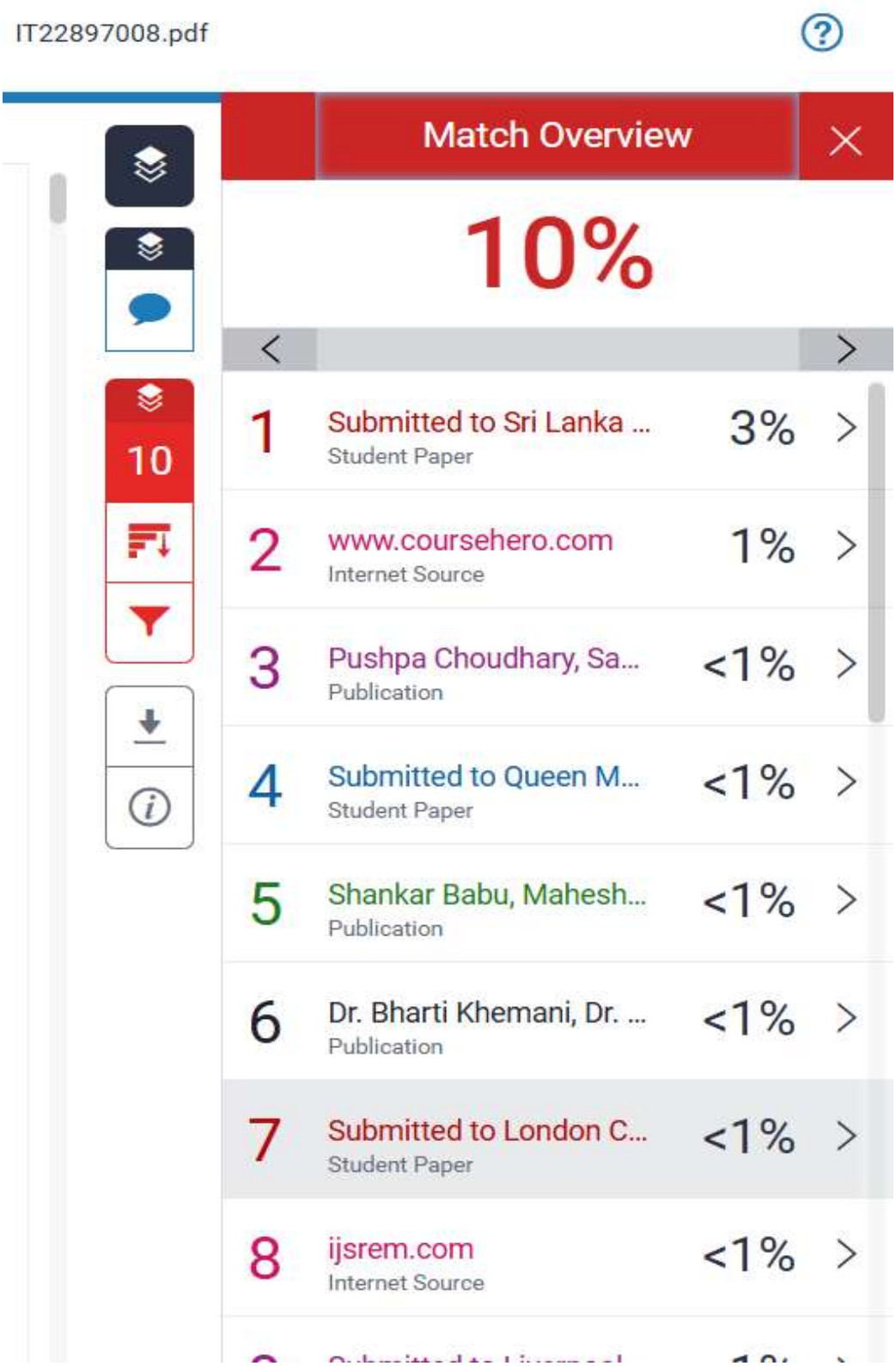
APPENDICES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